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| <p>Scope and goal of the project</p>                                         | <p>iNSpire AI, the project involves building an AI-enabled Software Development Life Cycle (SDLC) platform for Verizon's Network Systems team. It supports solution architects, system architects, business owners, and product owners by providing services such as user story generation, feature and MVP creation, code review, knowledge retrieval, and knowledge base validation. The platform leverages Gemini AI models and full-stack solutions to enhance and automate key development processes.</p> <p>The goal is to accelerate product delivery by reducing manual effort and time from requirements gathering to deployment. By generating AI-driven, domain-specific content for SME validation, the platform improves development speed, efficiency, and scalability. While currently serving the Network Systems team, it is designed for future expansion across Verizon's business and consumer groups.</p>                                                                                                                                                   |
| <p><b>Position Description and Detailed Job duties that equals 100 %</b></p> | <p>The Senior Data Scientist will support Verizon's Network Systems team by designing and developing AI-powered services to accelerate software development processes. Responsibilities include building full-stack GenAI solutions leveraging Gemini-based models, Python, Flask, and LangChain frameworks, integrating APIs, and deploying services on on-premise GPU servers. The role involves close collaboration with product owners, solution architects, SMEs, and peer developers to gather domain-specific requirements and deliver end-to-end solutions, including user story generation, knowledge retrieval, and code review automation. The Senior Data Scientist contributes to improving product delivery speed, reducing manual intervention, and supporting the scalability of the platform across Verizon business units. This position operates in a hybrid work environment with both remote and on-site responsibilities.</p> <p>Job responsibilities:</p> <p>Design and develop GenAI-powered backend services for user story generation, MVP feature</p> |





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|  | <p>creation, code review automation, and knowledge base validation.</p> <ul style="list-style-type: none"><li>• Integrate Gemini AI models, internal APIs, and LangChain frameworks into scalable Python-Flask-based solutions deployed on on-premise GPU servers.</li><li>• Collaborate with UI developers to define and implement API contracts, ensuring seamless backend-to-frontend integration for AI-driven applications.</li><li>• Perform prompt engineering, SQL database interactions, log processing, and generation of domain-specific reports and metrics to support solution validation.</li><li>• Collaborate with solution architects, data architects, SMEs, and cross-functional team members to gather business requirements, review architectural designs, and refine AI outputs.</li><li>• Build and maintain knowledge retrieval and validation pipelines, ensuring the integrity and contextual accuracy of AI-generated content.</li><li>• Conduct code reviews, enforce coding standards, and provide guidance to junior data scientists on solution design and task execution.</li><li>• Support platform scalability initiatives, enabling service expansion across multiple Verizon business and consumer verticals.</li><li>• Develop and maintain technical API documentation to support internal teams with service consumption, deployment, and integration.</li></ul> |
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|  | <p><b>Design and Develop AI-Powered Backend Services (30%)</b></p> <p>Craft robust and scalable backend services utilizing Python, Flask, and LangChain to automate key software development processes such as user story creation, knowledge base access, and code review workflows.</p> <p>Architect and implement modular code that encourages reusability, maintainability, and extensibility, ensuring a solid foundation for future enhancements and integrations.</p> <p>Translate complex business needs into technical designs, selecting optimal algorithms and models to achieve desired levels of accuracy, performance, and efficiency in the AI-powered services.</p> <p><b>Integrate AI Models, APIs, and Deploy Full-Stack Solutions (20%)</b></p> <p>Seamlessly integrate Gemini AI models with internal APIs to unlock AI capabilities within existing systems and create data-driven solutions.</p> <p>Deploy Python-Flask-based services on on-premise GPU servers, ensuring optimized performance and minimal latency to meet critical response time requirements.</p> <p>Collaborate closely with UI developers, defining clear API contracts and ensuring smooth end-to-end integration for a cohesive user experience across AI-driven applications.</p> <p><b>Validate AI Service Outputs through Prompt Engineering and Data Processing (20%)</b></p> <p>Employ advanced prompt engineering techniques to fine-tune the behavior of Gemini AI models, ensuring accurate, relevant, and contextually appropriate outputs for diverse tasks.</p> <p>Perform SQL database querying and log processing to extract valuable information for model</p> |
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|  | <p>validation, generate comprehensive reports, and provide actionable insights to stakeholders.</p> <p>Analyze AI service outputs to identify potential biases, inaccuracies, or performance bottlenecks, implementing iterative improvements to enhance reliability and effectiveness.</p> <p><b>Collaborate with Stakeholders to Refine AI-Driven Solutions (15%)</b></p> <p>Work closely with product owners and solution architects to fully understand business objectives and translate those into detailed specifications of AI-driven features and solutions.</p> <p>Engage with subject matter experts (SMEs) to gain domain-specific knowledge, ensuring that AI models are trained and optimized to address real-world challenges and opportunities.</p> <p>Actively participate in design reviews, providing valuable feedback to shape the architecture and ensure that solutions align with the overall strategic vision and technical requirements.</p> <p><b>Code Reviews, Mentorship, and Platform Scalability (15%)</b></p> <p>Conduct thorough code reviews to enforce coding standards and ensure the overall quality, maintainability, and security of AI-powered solutions.</p> <p>Mentor junior data scientists, providing guidance on solution design, task execution, and best practices to foster a culture of continuous learning and professional development.</p> <p>Contribute to platform scalability initiatives by designing services that can be efficiently deployed across multiple Verizon business units and consumer verticals, ensuring seamless integration and consistent performance.</p> |
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## Software used

Programming Languages and Frameworks:  
Python, Flask, SQL

AI/ML Tools and Libraries:  
LangChain, Llama Index, HuggingFace  
Transformers, Scikit-learn, TensorFlow,  
PyTorch, Natural Language Processing (NLP)  
libraries

Database and Storage Technologies:  
PostgreSQL, AWS S3

DevOps, Deployment, and Infrastructure Tools:  
Docker, Kubernetes, Jenkins, GitLab

Cloud Platforms:  
Amazon Web Services (AWS)

Documentation and Collaboration Tools:  
JIRA, Confluence

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